

Varieties of degree 3 in PG(2,4)

Orbiter

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$$q = 4$$

$$n = 2$$

$$\text{degree of poly} = 3$$

The Varieties of degree 3 in $PG(2, 4)$, summary

Objects with 13 Points

There are 1 objects with 13 Points:

orbit : rep : go : poly : Pts

4 : (11) : 288 : $11=X^3+Y^3$: "2, 3, 4, 5, 6, 11, 12, 14, 15, 16, 18, 19, 20"

Objects with 12 Points

There are 1 objects with 12 Points:

orbit : rep : go : poly : Pts

2 : (9) : 54 : $9=XYZ$: "0, 1, 2, 4, 5, 6, 7, 8, 9, 10, 13, 17"

Objects with 10 Points

There are 1 objects with 10 Points:

orbit : rep : go : poly : Pts

31 : (87743) : 10 : $87743=X^3+2Y^3+2Z^3+X^2Y+X^2Z+XYZ$: "3, 5, 8, 10, 13, 14, 15, 17, 18, 20"

Objects with 9 Points

There are 4 objects with 9 Points:

orbit : rep : go : poly : Pts

1 : (3) : 144 : $3=X^2Y$: "0, 1, 2, 7, 8, 9, 10, 13, 17"

6 : (18) : 216 : $18=X^3+Y^3+Z^3$: "4, 5, 6, 7, 8, 9, 10, 13, 17"

9 : (24) : 27 : $24=3X^3+2Y^3+Z^3$: "3, 11, 12, 14, 15, 16, 18, 19, 20"

15 : (175) : 12 : $175=Y^3+Z^3+X^2Y+X^2Z$: "0, 3, 4, 7, 10, 11, 12, 13, 17"

Objects with 8 Points

There are 2 objects with 8 Points:

orbit : rep : go : poly : Pts

19 : (87383) : 6 : $87383=X^3+XYZ$: "1, 2, 3, 10, 13, 16, 17, 19"

24 : (87466) : 2 : $87466=Y^3+Z^3+X^2Y+XYZ$: "0, 3, 4, 10, 13, 14, 17, 18"

Objects with 7 Points

There are 2 objects with 7 Points:

orbit : rep : go : poly : Pts

12 : (50) : 6 : $50=2X^3+Y^3+Z^3+X^2Y$: "5, 10, 12, 13, 14, 17, 19"

13 : (51) : 6 : $51=3X^3+Y^3+Z^3+X^2Y$: "6, 10, 11, 13, 16, 17, 18"

Objects with 6 Points

There are 6 objects with 6 Points:

orbit : rep : go : poly : Pts

21 : (87404) : 6 : $87404=2X^3+Y^3+Z^3+XYZ$: "10, 11, 13, 14, 17, 20"

22 : (87405) : 6 : $87405=3X^3+Y^3+Z^3+XYZ$: "10, 12, 13, 15, 17, 18"

25 : (87468) : 3 : $87468=2X^3+Y^3+Z^3+X^2Y+XYZ$: "5, 10, 13, 15, 16, 17"

26 : (87469) : 3 : $87469=3X^3+Y^3+Z^3+X^2Y+XYZ$: "6, 10, 13, 17, 19, 20"

28 : (87471) : 2 : $87471=X^3+2Y^3+Z^3+X^2Y+XYZ$: "5, 7, 8, 9, 15, 16"

32 : (91990) : 30 : $91990=2X^2Z+Y^2Z+XYZ$: "0, 1, 2, 4, 5, 6"

Objects with 5 Points

There are 4 objects with 5 Points:

orbit : rep : go : poly : Pts

0 : (0) : 2880 : $0=X^3$: "1, 2, 10, 13, 17"

10 : (34) : 96 : $34=2X^3+Y^3+X^2Y$: "2, 5, 11, 16, 18"

11 : (44) : 3 : $44=Z^3+X^2Y$: "0, 1, 3, 15, 20"

16 : (176) : 4 : $176=X^3+Y^3+Z^3+X^2Y+X^2Z$: "10, 13, 16, 17, 19"

Objects with 4 Points

There are 4 objects with 4 Points:

orbit : rep : go : poly : Pts

3 : (10) : 6 : $10=X^3+Y^3+Z^3+X^2Y+X^2Z+XY^2+Y^2Z+XZ^2+YZ^2+XYZ$: "3, 4, 7, 10"

20 : (87388) : 3 : $87388=2X^3+Y^3+XYZ$: "2, 12, 15, 18"

27 : (87470) : 2 : $87470=2Y^3+Z^3+X^2Y+XYZ$: "0, 6, 19, 20"

30 : (87474) : 2 : $87474=3Y^3+Z^3+X^2Y+XYZ$: "0, 5, 15, 16"

Objects with 3 Points

There are 4 objects with 3 Points:

orbit : rep : go : poly : Pts

7 : (19) : 18 : 19= $2X^3 + Y^3 + Z^3$: "10, 13, 17"

8 : (20) : 18 : 20= $3X^3 + Y^3 + Z^3$: "10, 13, 17"

17 : (180) : 9 : 180= $X^3 + 2Y^3 + Z^3 + X^2Y + X^2Z$: "5, 15, 18"

18 : (181) : 9 : 181= $2X^3 + 2Y^3 + Z^3 + X^2Y + X^2Z$: "8, 16, 19"

Objects with 2 Points

There are 1 objects with 2 Points:

orbit : rep : go : poly : Pts

29 : (87472) : 2 : 87472= $2X^3 + 2Y^3 + Z^3 + X^2Y + XYZ$: "11, 12"

Objects with 1 Points

There are 2 objects with 1 Points:

orbit : rep : go : poly : Pts

5 : (12) : 144 : 12= $2X^3 + Y^3$: "2"

14 : (55) : 24 : 55= $3X^3 + 2Y^3 + Z^3 + X^2Y$: "4"

Objects with 0 Points

There are 1 objects with 0 Points:

orbit : rep : go : poly : Pts

23 : (87409) : 63 : 87409= $3X^3 + 2Y^3 + Z^3 + XYZ$: ""